INTRODUCTION TO FLUID MECHANICS

Department of Chemical Engineering

CHEM ENG 2O04
Course Notes

THE BERNOULLI EQUATION – Part I

Fox & McDonald

10th & 9th editions: 4.1, 4.8, 6.3, 8.6 – 8.8

PIPE-FLO

Tutorial #4 'Simulation Calculations'

“There are three ways to ultimate success:

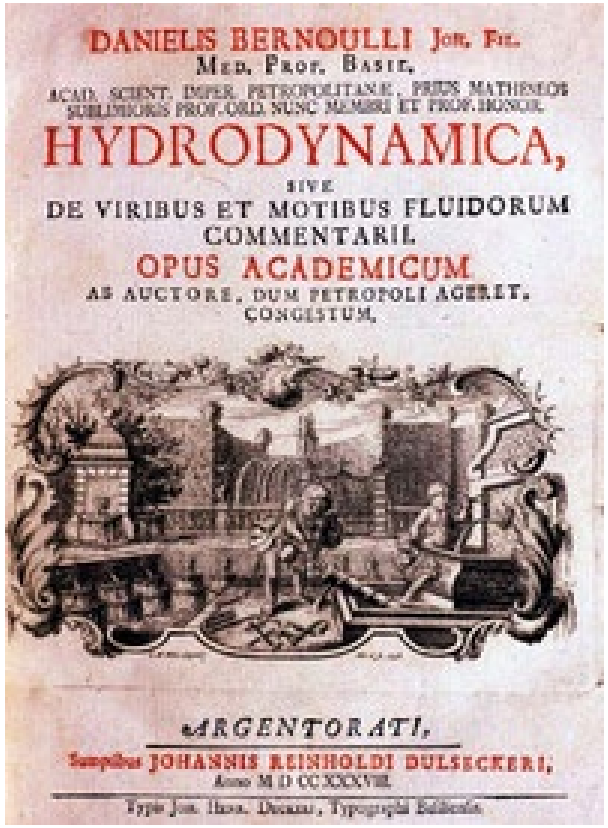
”

Fred Rogers

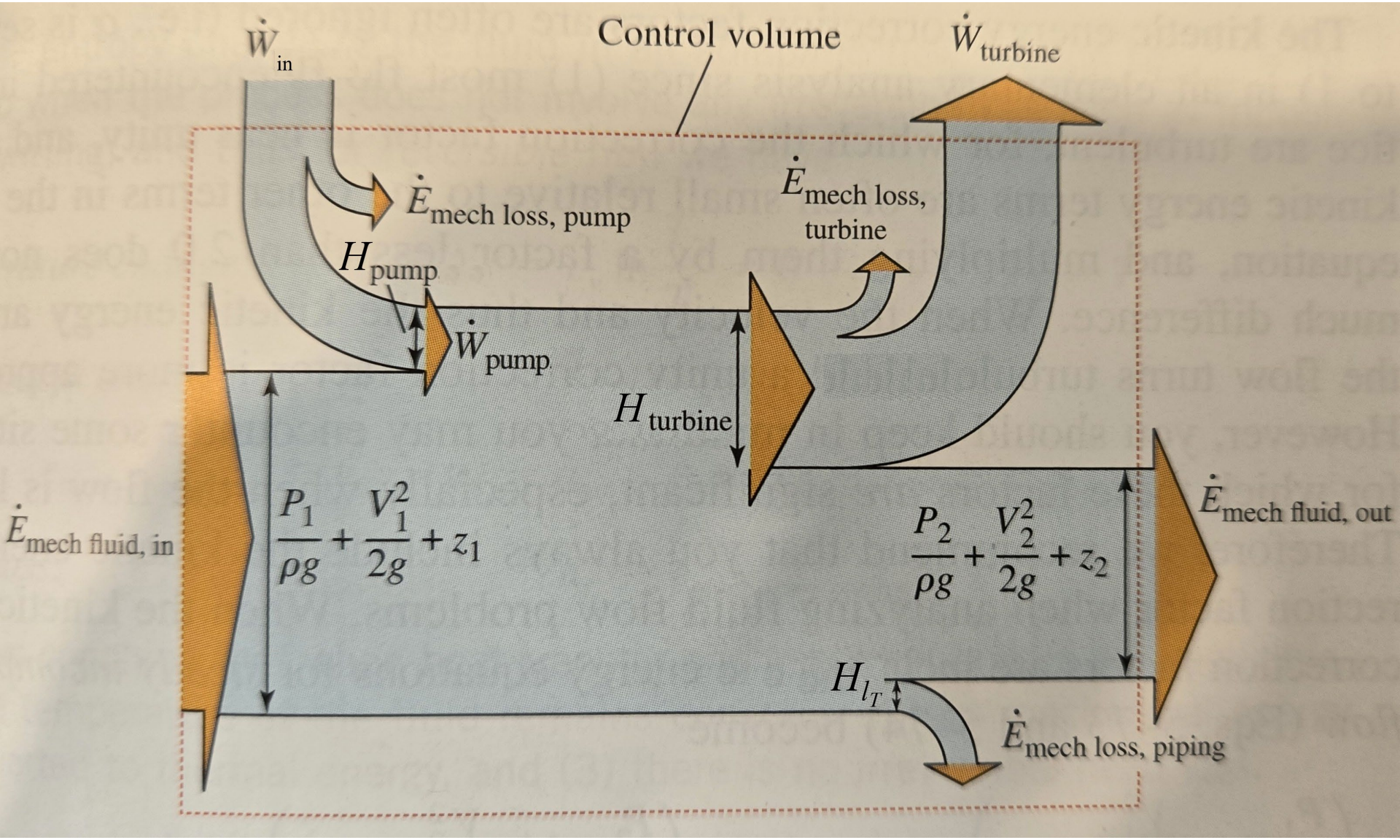
BERNOULLI EQUATION

aka "Total Mechanical Energy Balance"
"Mechanical Energy Equation"

Daniel Bernoulli's book published in 1738:
"Steady, one-dimensional, incompressible, frictionless flow without heat transfer or shear work"


DANIELIS BERNOULLI Jon. Fil.
Med. Prof. Basil.
ACAD. SCIENT. IMPER. PETROPOLITANÆ, PRIUS MATHESEOS SUBLIMIORIS PROF. ORD. NUNC MEMBRI ET PROF. HONOR.
HYDRODYNAMICA,
SIVE
DE VIRIBUS ET MOTIBUS FLUIDORUM COMMENTARII.
OPUS ACADEMICUM
AB AUCTORE, DUM PETROPOLI AGERET, CONGESTUM.
ARGENTORATI,
Sumptibus JOHANNIS REINHOLDI DULSECKERI,
Anno M D CC XXXVIII.
Typis Jon. Henr. Decani, Typographi Solitarii.

Sankey diagram of the full Bernoulli Equation



DERIVATION OF BERNOULLI EQUATION

Conservation of
Linear Momentum

Energy Balance
1st Law of Thermodynamics
Eqn 6.15

Force Balance on
Control Volume
Fox Chap 4.4
Eqn 4.24

Navier-Stokes
Equations
Fox Chap 6.2 & 6.3
Eqn 6.8

1st Law of Thermodynamics

Rate of heat transfer

Rate of work

- Conservation of energy for a system: $$\dot{Q} - \dot{W} = \left.\frac{dE}{dt}\right|_{system}$$ Eqn 4.4a

Defined as "work done by the system on its surroundings"

1st Law of Thermodynamics

Rate of heat transfer

Rate of work

- Conservation of energy for a system: $$\dot{Q} - \dot{W} = \left.\frac{dE}{dt}\right|_{system}$$ Eqn 4.4a

- The total energy (E) is given by: $$E = \int_{\forall(system)} e\rho d\forall$$ Eqn 4.4b

where $$e = u + \frac{V^2}{2} + gz$$ Eqn 4.4c

Internal energy per unit mass

kinetic energy per unit mass

potential energy per unit mass

- Use Eqn 4.53 to relate rate of change of E of system to formulation for use with a control volume (CV) with control surface (CS):

$$\left.\frac{dE}{dt}\right|_{system} = \frac{\partial}{\partial t}\int_{CV} e\rho d\forall + \int_{CS} e\rho \overline{V} \bullet d\overline{A}$$ Eqn 4.53

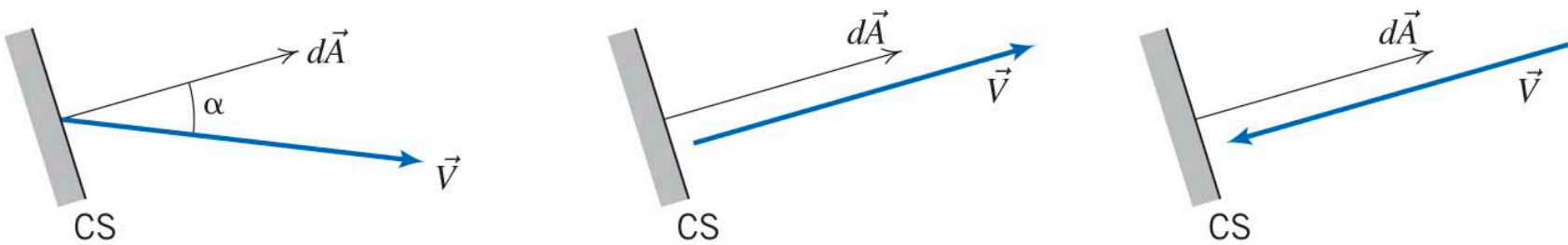
1st Law of Thermodynamics

$$\dot{Q} - \dot{W} = \underbrace{\frac{\partial}{\partial t}\int_{CV} e\rho d\forall}_{\text{Rate of change of energy in control volume}} + \underbrace{\int_{CS} e\rho \overline{V} \bullet d\overline{A}}_{\text{Rate at which energy is entering/leaving the control surface}}$$

Eqn 4.54

- Define a vector area element $d\overline{A}$ of the surface with magnitude dA and direction always outward normal of the area element

  Dot product (aka scalar product) automatically distinguishes inflows and outflows



$\vec{V} \cdot d\vec{A} = VdA \cos \alpha$ (a) General inlet/exit

$\vec{V} \cdot d\vec{A} = +VdA$ (b) Normal exit

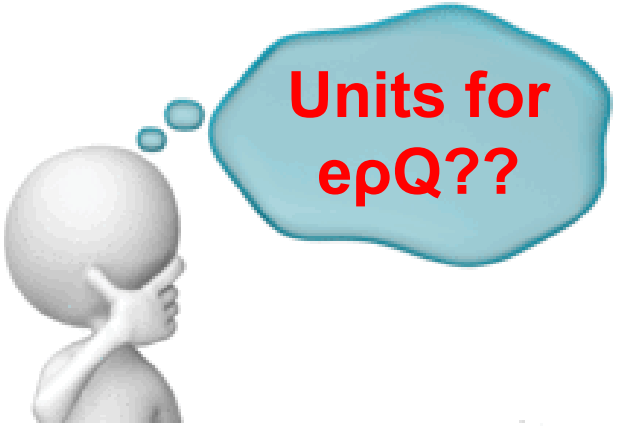
$\vec{V} \cdot d\vec{A} = -VdA$ (c) Normal inlet

Figure 4.3

1st Law of Thermodynamics

$$\dot{Q} - \dot{W} = \underbrace{\frac{\partial}{\partial t}\int_{CV} e\rho d\forall}_{\text{Rate of change of energy in control volume}} + \underbrace{\int_{CS} e\rho \overline{V} \bullet d\overline{A}}_{\text{Rate at which energy is entering/leaving the control surface}}$$

Eqn 4.54



1st Law of Thermodynamics

$$\dot{Q} - \dot{W} = \underbrace{\frac{\partial}{\partial t}\int_{CV} e\rho d\forall}_{\text{Rate of change of energy in control volume}} + \underbrace{\int_{CS} e\rho \overline{V} \bullet d\overline{A}}_{\text{Rate at which energy is entering/leaving the control surface}}$$

Eqn 4.54

- Work term: $\dot{W} = \underbrace{\dot{W}_{flow}}_{\text{due to pressure distribution}} + \dot{W}_{shaft}$

MUST be absolute pressure!!!

$$\dot{W}_{flow} = \int_{CS} \left(\frac{P}{\rho} \right) \rho \overline{V} \bullet d\overline{A}$$

Rate of 'flow work'

$$\text{work} = \text{force} \times \text{distance}$$

$$\frac{\text{work}}{\text{time}} = \left(\frac{\text{pressure}}{\text{density}}\right)(\text{mass flow rate})$$

1st Law of Thermodynamics

$$\dot{Q} - \dot{W} = \underbrace{\frac{\partial}{\partial t}\int_{CV} e\rho d\forall}_{\text{Rate of change of energy in control volume}} + \underbrace{\int_{CS} e\rho \overline{V} \bullet d\overline{A}}_{\text{Rate at which energy is entering/leaving the control surface}}$$

Eqn 4.54

- Work term: $\dot{W} = \underbrace{\dot{W}_{flow}}_{\text{due to pressure distribution}} + \dot{W}_{shaft}$

$$\dot{W}_{flow} = \int_{CS}\left(\frac{P}{\rho}\right)\rho\overline{V} \bullet d\overline{A}$$

- Combining the terms gives:

$$\dot{Q} - \int_{CS}\left(\frac{P}{\rho}\right)\rho\overline{V} \bullet d\overline{A} - \dot{W}_{shaft} = \frac{\partial}{\partial t}\int_{CV} e\rho d\forall + \int_{CS}\left(u + \frac{V^2}{2} + gz\right)\rho\overline{V} \bullet d\overline{A}$$

$$\dot{Q} - \dot{W}_{shaft} = \frac{\partial}{\partial t}\int_{CV} e\rho d\forall + \int_{CS}\left(u + \frac{V^2}{2} + gz + \frac{P}{\rho}\right)\rho\overline{V} \bullet d\overline{A}$$

Eqn 4.56

1st Law of Thermodynamics

$$\dot{Q} - \dot{W}_{shaft} = \frac{\partial}{\partial t}\int_{CV} e\rho d\forall + \int_{CS}\left(u + \frac{V^2}{2} + gz + \frac{P}{\rho}\right)\rho\bar{V}\bullet d\bar{A} \quad \text{Eqn 4.56}$$

- *Assumption #1* → Steady-flow $\frac{\partial}{\partial t}\ldots = 0$
- *Assumption #2* → Flow crosses the control surface via single inlet (1) and single outlet (2) with average velocities V_1 and V_2

$$\dot{Q} - \dot{W}_{shaft} = \left(u_1 + \alpha_1\frac{V_1^2}{2} + gz_1 + \frac{P_1}{\rho_1}\right)\left(-\rho_1 V_1 A_1\right) + \left(u_2 + \alpha_2\frac{V_2^2}{2} + gz_2 + \frac{P_2}{\rho_2}\right)\left(+\rho_2 V_2 A_2\right)$$

Internal energy per unit mass | kinetic energy per unit mass | potential energy per unit mass | flow energy per unit mass

$\alpha \equiv$ kinetic energy coefficient (see Eqn 8.26b); a correction factor that is required since we use average velocities to calculate kinetic energy

- For laminar flow $\alpha = 2$
- For turbulent flow $\alpha \approx 1$

1st Law of Thermodynamics

- By conservation of mass: $\dot{m} = \rho_1 V_1 A_1 = \rho_2 V_2 A_2$

$$0 = \dot{m}\left[\left(\frac{P_2}{\rho_2} + \alpha_2 \frac{V_2^2}{2} + gz_2\right) - \left(\frac{P_1}{\rho_1} + \alpha_1 \frac{V_1^2}{2} + gz_1\right)\right] + \dot{m}(u_2 - u_1) - \dot{Q} + \dot{W}_{shaft}$$

- *Assumption #3* → No shaft work (i.e. pumps, turbines) occurs
- *Assumption #4* → Incompressible fluid ($\rho_1 = \rho_2 = \rho$)

$$0 = \left(\frac{P_2}{\rho} + \alpha_2 \frac{V_2^2}{2} + gz_2\right) - \left(\frac{P_1}{\rho} + \alpha_1 \frac{V_1^2}{2} + gz_1\right) + \left(u_2 - u_1 - \frac{\dot{Q}}{\dot{m}}\right) \qquad \text{Eqn 6.14}$$

- *Assumption #5* → Frictionless flow; $u_2 - u_1 - \frac{\dot{Q}}{\dot{m}} = 0$

$$0 = \left(\frac{P_2}{\rho} + \alpha_2 \frac{V_2^2}{2} + gz_2\right) - \left(\frac{P_1}{\rho} + \alpha_1 \frac{V_1^2}{2} + gz_1\right)$$

Finally….

$$\boxed{\frac{P_1}{\rho} + \alpha_1 \frac{V_1^2}{2} + gz_1 = \frac{P_2}{\rho} + \alpha_2 \frac{V_2^2}{2} + gz_2}$$

Bernoulli Equation

Eqn 6.15

Bernoulli Equation

$$\frac{P_1}{\rho} + \alpha_1 \frac{V_1^2}{2} + gz_1 = \frac{P_2}{\rho} + \alpha_2 \frac{V_2^2}{2} + gz_2$$

"Energy form" of Bernoulli Equation since all terms have units of energy/mass

or alternatively

$$\frac{P_1}{\rho g} + \alpha_1 \frac{V_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \alpha_2 \frac{V_2^2}{2g} + z_2$$

"Head form" of Bernoulli Equation since all terms have units of length

Relates (absolute) pressure , velocity, and elevation at two points along a streamline in a fluid in steady-state flow

BUT be careful:

"The most famous, and ABUSED, equation in all of fluid mechanics. It is always tempting to use because it is a simple algebraic equation for relating the pressure, velocity, and elevation in a fluid.....you should keep the restrictions firmly in mind whenever you consider using the Bernoulli equation!"

$$\boxed{\frac{P}{\rho} + \alpha \frac{V^2}{2} + gz = \text{constant}}$$

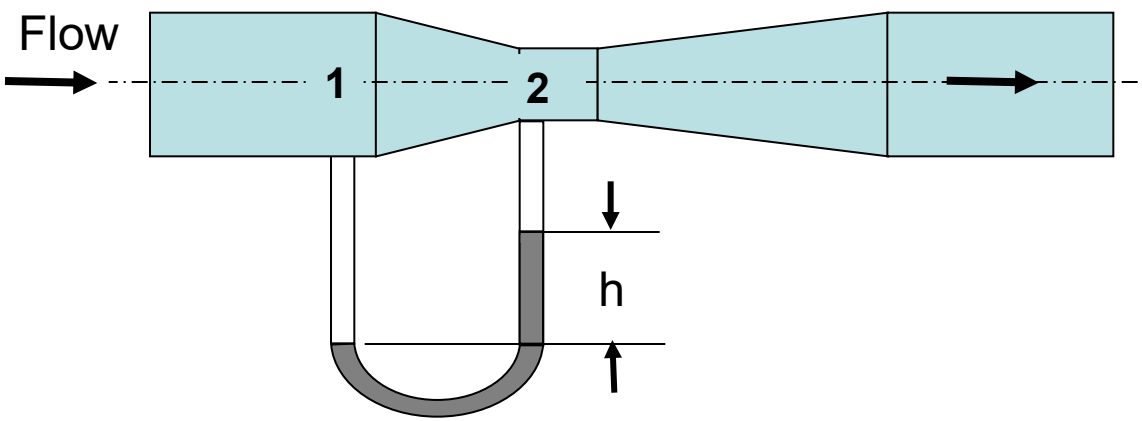
Eqn 4.24
Eqn 6.8
Eqn 6.15

Del operator

$$\nabla\left(\frac{P}{\rho} + \alpha \frac{V^2}{2} + gz\right) = 0$$

Relates (absolute) pressure, velocity, and elevation at two points along a **streamline** in a fluid in steady-state flow

Restriction flow meters (e.g. Venturi meter)



$$Q = \frac{\pi}{4} D_2^2 \sqrt{\frac{2 \cdot (P_1 - P_2)}{\rho \left(1 - \frac{D_2^4}{D_1^4}\right)}}$$

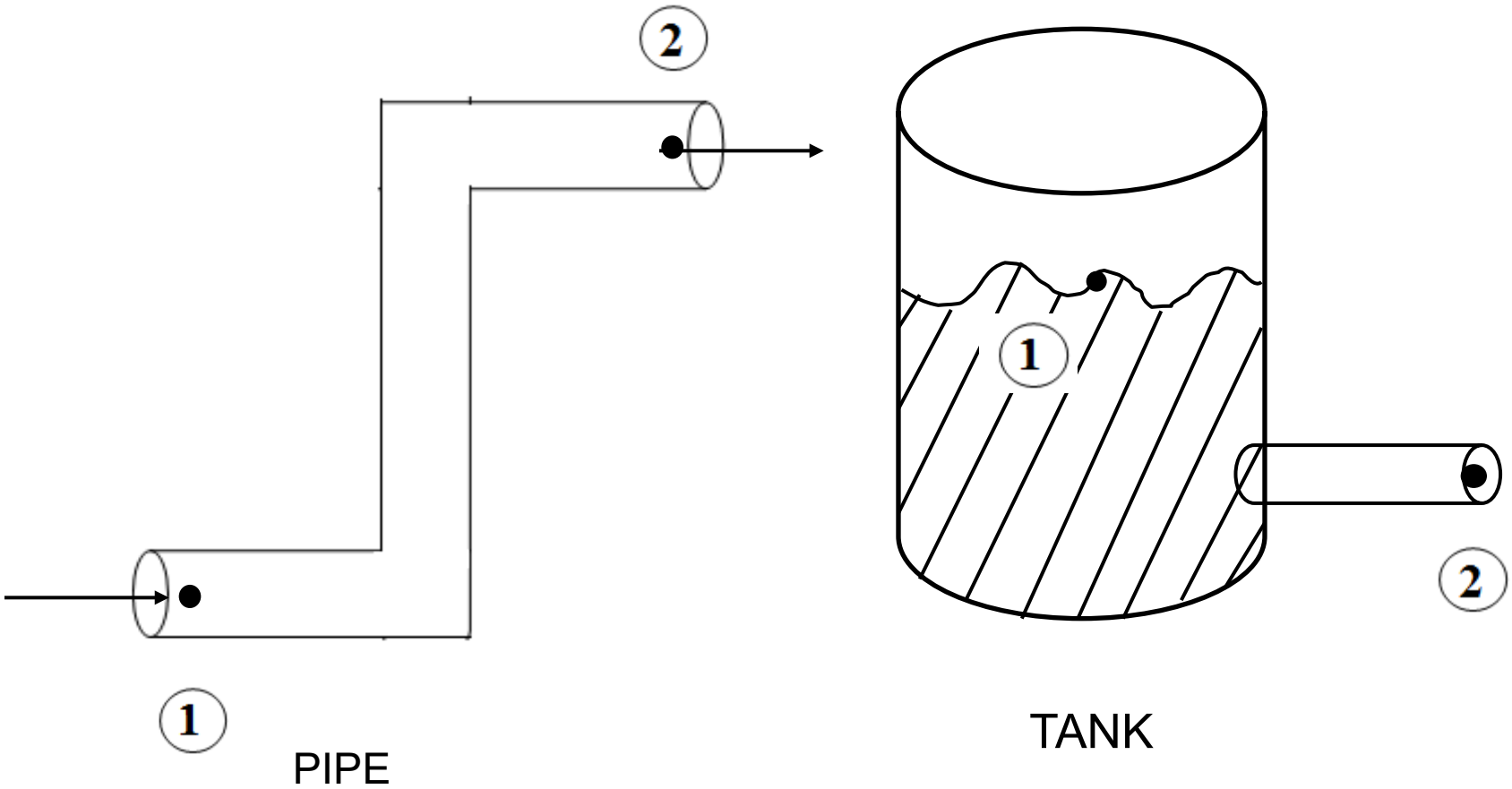
Derived from mass balance and mechanical energy balance (i.e. Bernoulli equation) – will come back to this later in the course!

- The difference in height between two legs of manometer is proportional to (P_1-P_2)
- Disadvantage is that the measured output (P_1-P_2) is not linear with flow rate (Q)!

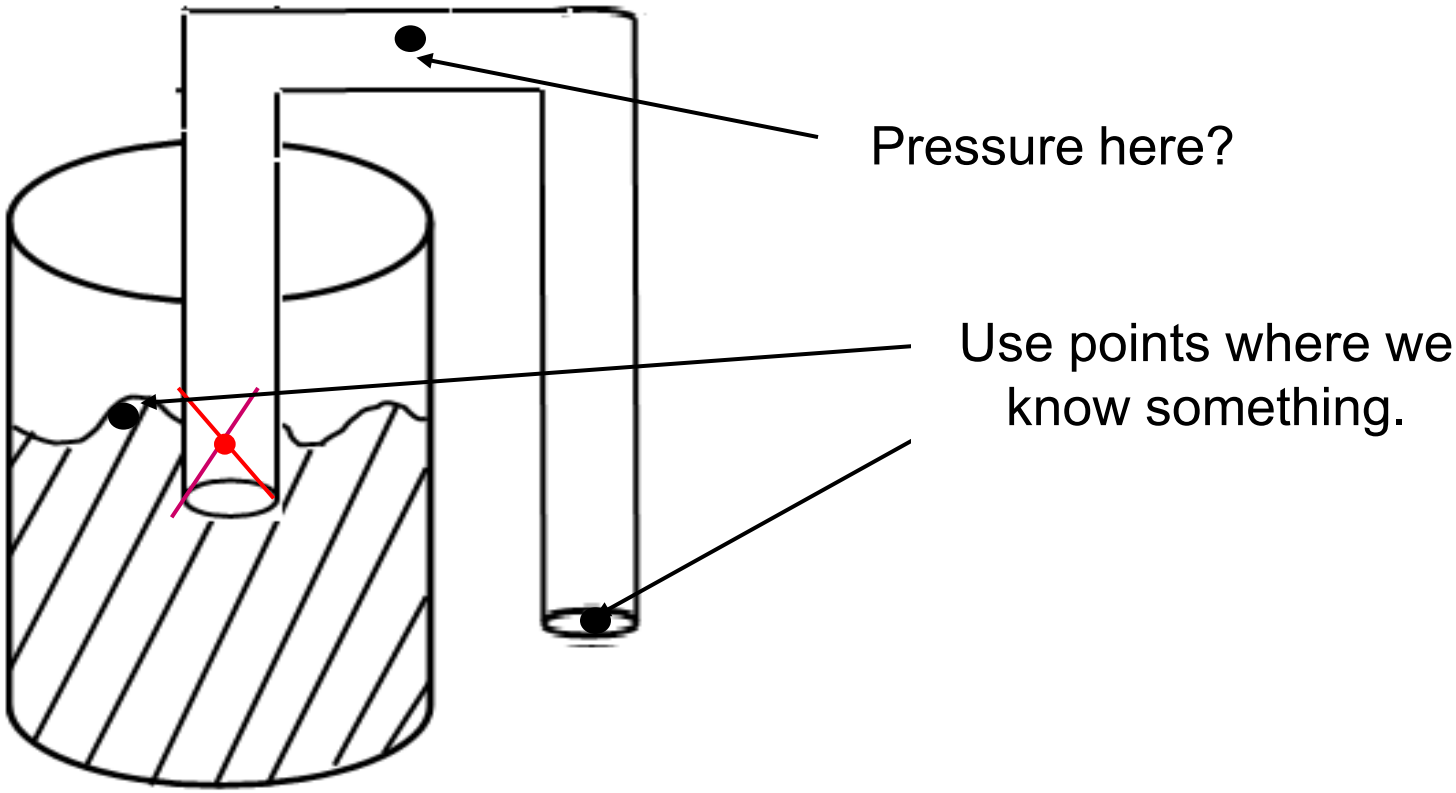
TIPS FOR USING BERNOULLI EQUATION

1. Determine the system and what is the start and end point over which Bernoulli Equation is applied.

Convention for this course is that Location 1 is ALWAYS upstream of Location 2



2. Points 1 & 2 ideally should be assigned to locations where you know the most information

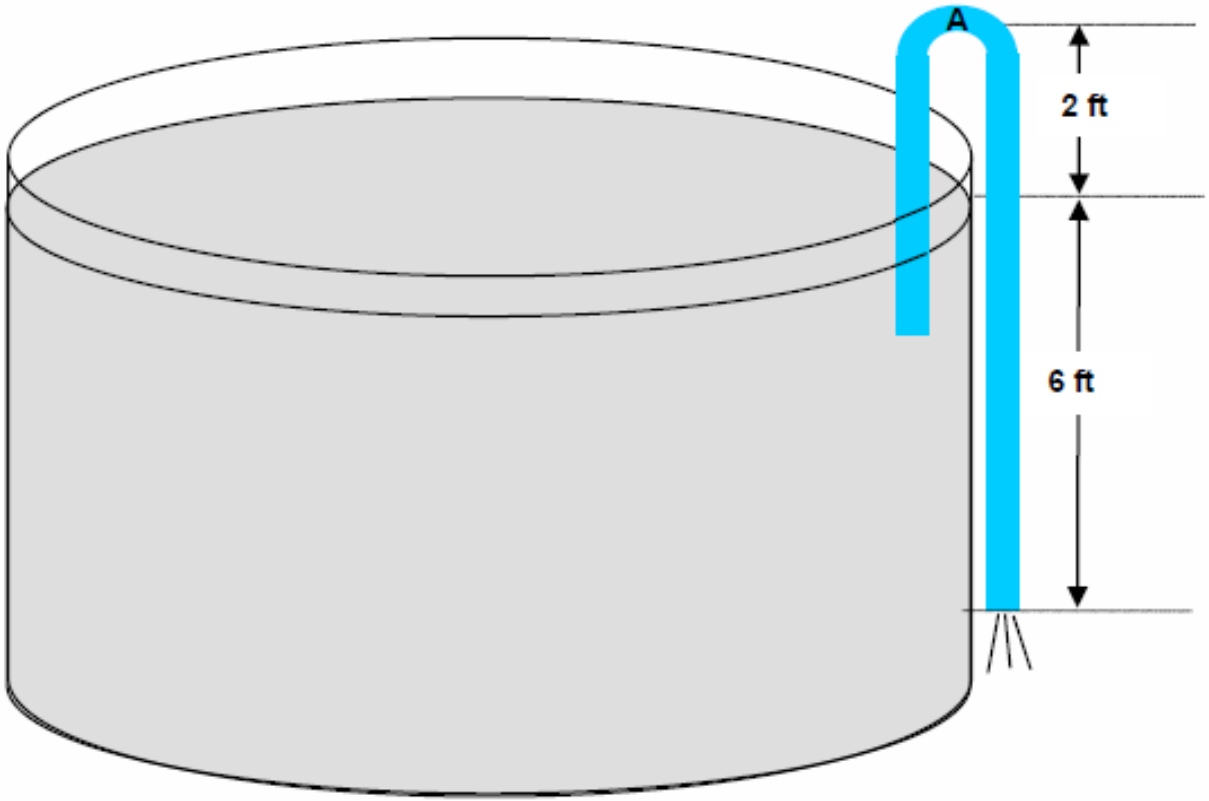


In-class problem

A plastic tube (internal diameter = 1”) is used as a siphon to drain a large reservoir of water that is open to the atmosphere. The bend in the tube is 2 ft above the reservoir surface; the siphon outlet is 6 ft below the reservoir surface. The water issues from the bottom of the siphon as a free jet at atmospheric pressure.

a) Select an appropriate control volume and reference points 1 and 2 to be used in the Bernoulli equation to determine the fluid velocity at the siphon outlet.

b) Select a new set of reference points 1 and 2 to determine the pressure at point A on the diagram. Report the value as both an absolute and gauge pressure.



Question #6 on Midterm #1

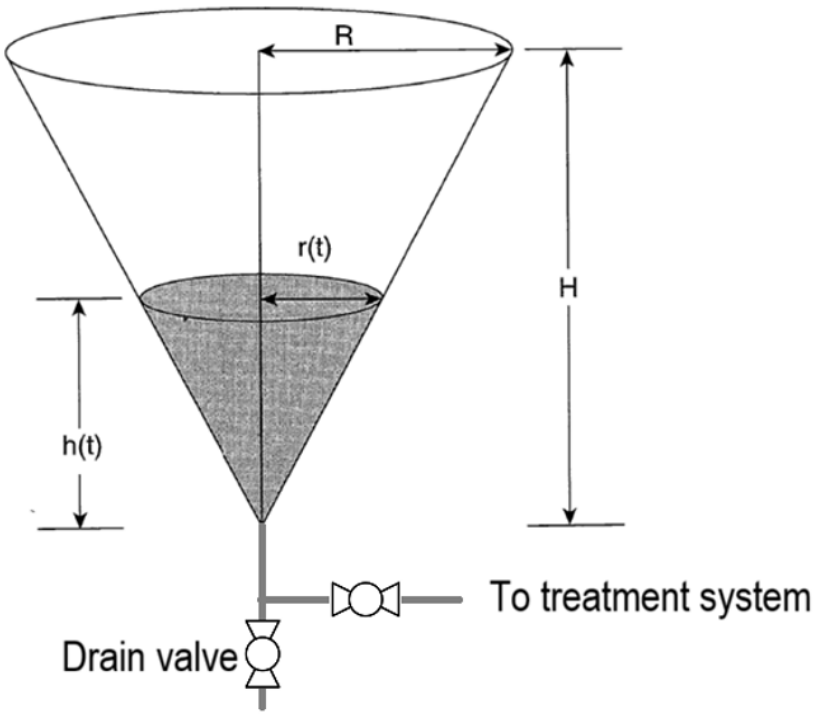
As mentioned in lecture, an inverted conical tank (with height H and radius R) with a bottom drain valve (with internal diameter d) is quite useful for lab-scale water treatment systems because the tank can be easily emptied by simply opening the drain valve. For a fully-opened drain valve, the average velocity of water leaving the tank can be approximated as

$\sqrt{2gh_{tank}}$

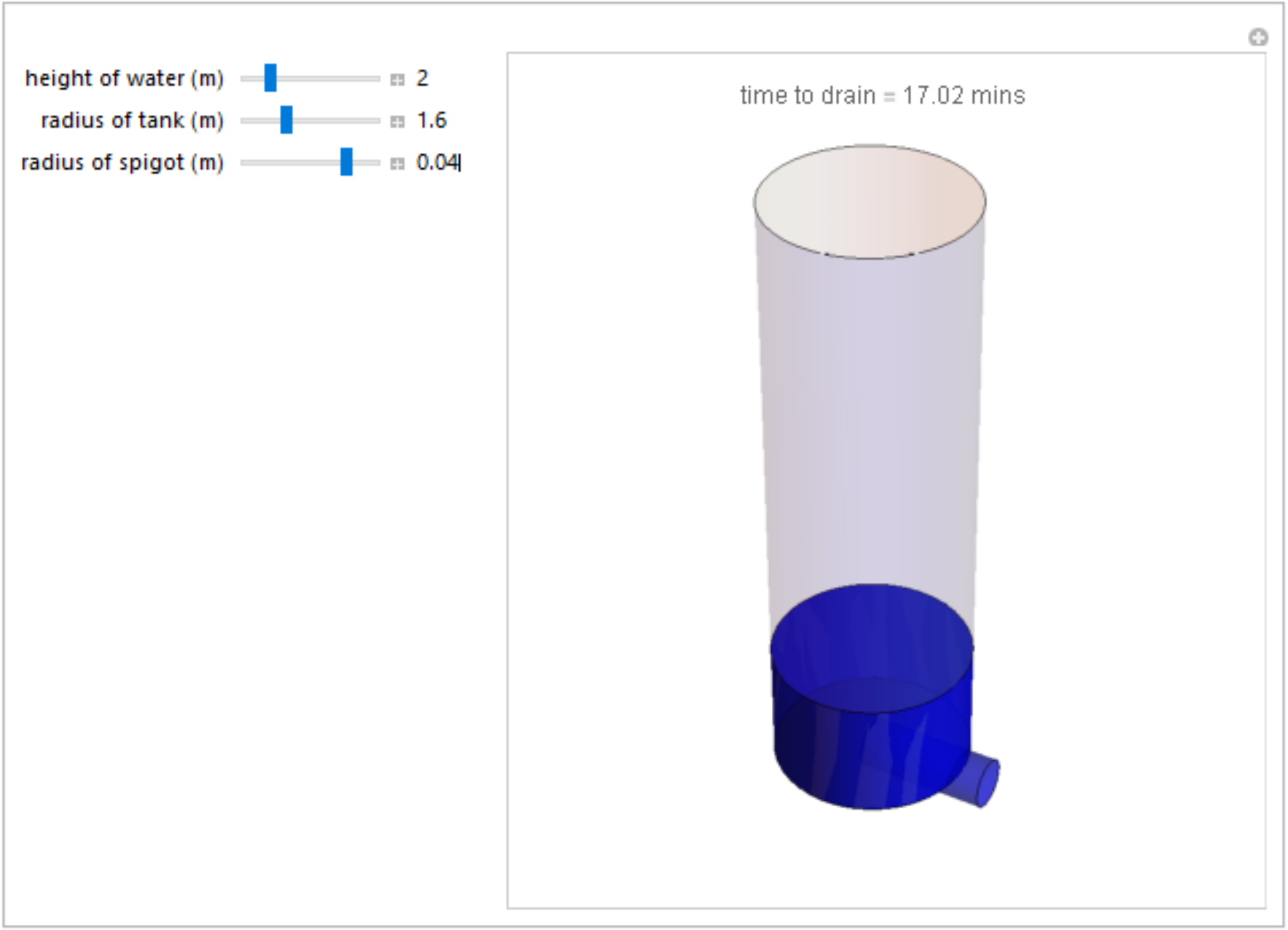
where

g is the gravitational constant (9.81 m/s^2)

h_{tank} is the height of water in the tank (m)



http://demonstrations.wolfram.com/TimeToDrainATankUsingTorricellisLaw/



- Review the following in textbook (both 10th & 9th):
  - Example 6.2: Pitot Tube
  - Example 6.3: Nozzle Flow
  - Example 6.4: Flow Through a Siphon

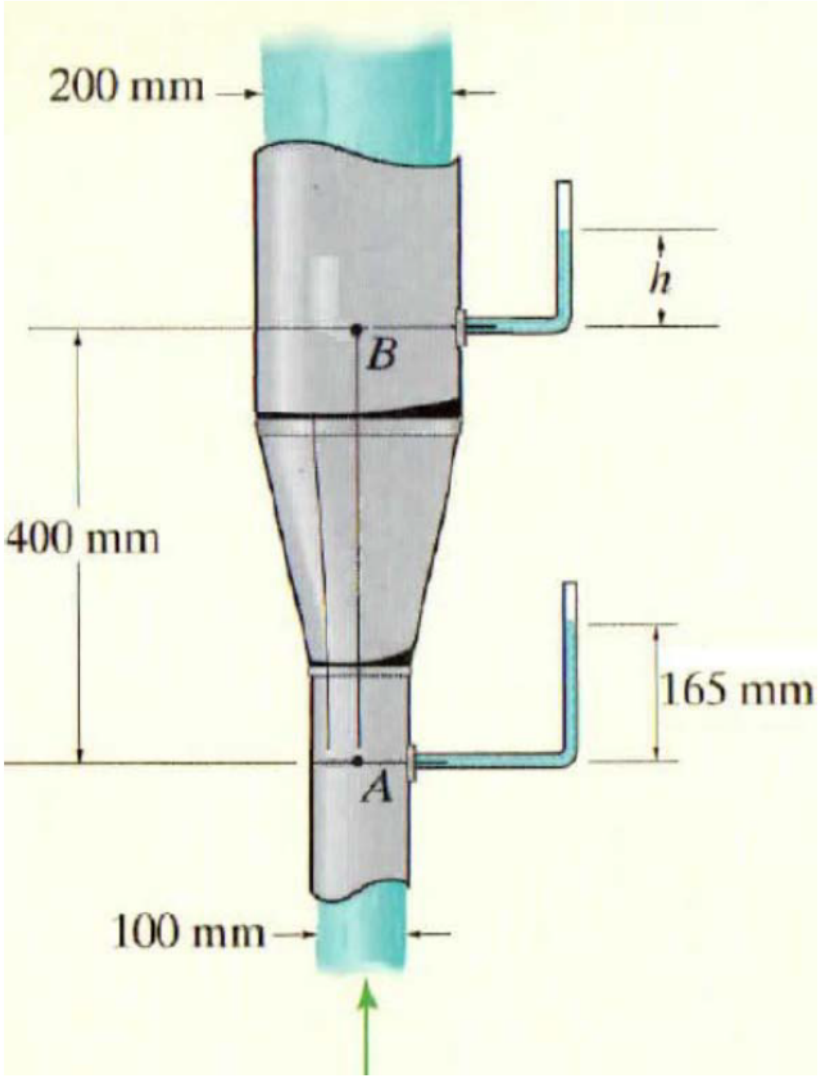
In-class problem

Cold water (at 4 °C) flows up through the vertical pipe that is connected to the expansion joint as shown in the figure. Two open-ended manometers are attached to the pipe on either side of the expansion to indicate the respective pressure values.

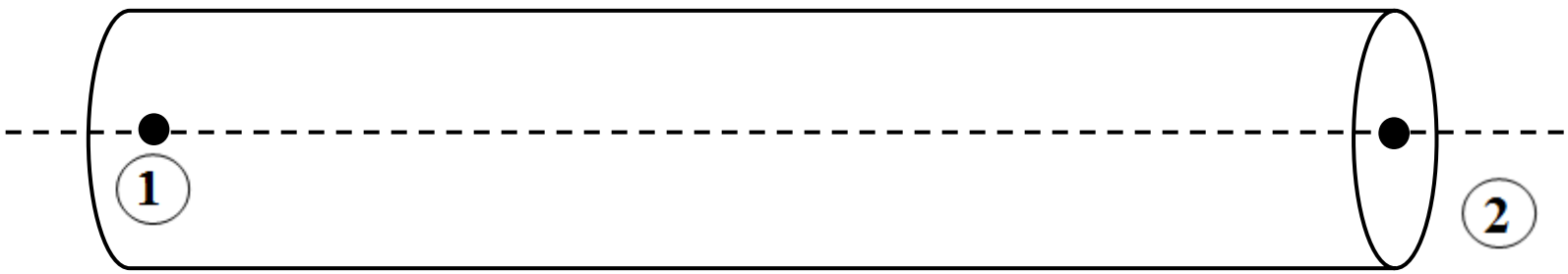
a. Write the correct form of the Bernoulli equation (in units of "energy per mass") that describes the flow along the streamline between points A and B.

b. If the volumetric flow rate of water through the pipe is 0.02 m^3/s, calculate the absolute pressure at point B.

c. Determine the height of fluid h (in mm) in the manometer at point B.



Let's try the following example:
Consider laminar flow through a horizontal pipe of constant diameter.



$$\frac{P_1}{\rho g} + \alpha_1 \frac{V_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \alpha_2 \frac{V_2^2}{2g} + z_2$$

Horizontal pipe →

Conservation of mass →

$$\boxed{\begin{aligned} P_1 &= P_2 \\ \Delta P &= 0 \end{aligned}}$$

WRONG!

Constant diameter →

Remember the assumptions in our derivation…

- By conservation of mass: $\dot{m} = \rho_1 V_1 A_1 = \rho_2 V_2 A_2$

$$0 = \dot{m}\left[\left(\frac{P_2}{\rho_2} + \alpha_2 \frac{V_2^2}{2} + gz_2\right) - \left(\frac{P_1}{\rho_1} + \alpha_1 \frac{V_1^2}{2} + gz_1\right)\right] + \dot{m}(u_2 - u_1) - \dot{Q} + \dot{W}_{shaft}$$

- *Assumption #3* → No shaft work (i.e. pumps, turbines) occurs
- *Assumption #4* → Incompressible fluid ($\rho_1 = \rho_2 = \rho$)

$$0 = \left(\frac{P_2}{\rho} + \alpha_2 \frac{V_2^2}{2} + gz_2\right) - \left(\frac{P_1}{\rho} + \alpha_1 \frac{V_1^2}{2} + gz_1\right) + \left(u_2 - u_1 - \frac{\dot{Q}}{\dot{m}}\right) \qquad \text{Eqn 6.14}$$

- ***Assumption #5*** → **Frictionless flow;** $\boxed{u_2 - u_1 - \frac{\dot{Q}}{\dot{m}} = 0}$

$$0 = \left(\frac{P_2}{\rho} + \alpha_2 \frac{V_2^2}{2} + gz_2\right) - \left(\frac{P_1}{\rho} + \alpha_1 \frac{V_1^2}{2} + gz_1\right)$$

h_{lT} accounts for total energy loss

Need to account for:

1. Irreversible conversion of mechanical energy at Location 1 to unwanted thermal energy
2. Loss of energy via heat transfer

$$h_{l_T} = h_l + h_{l_m}$$

Dependent on flow profile through the duct

Independent of pipe orientation

"Major" losses due to frictional effects

"Minor" losses due to entrances, fittings (elbows, valves), area changes, etc.

Bernoulli Equation WITH losses

Lower case h

Units = length^2/time^2 = energy/mass

$$\frac{P_1}{\rho} + \alpha_1 \frac{V_1^2}{2} + gz_1 = \frac{P_2}{\rho} + \alpha_2 \frac{V_2^2}{2} + gz_2 + h_{l_T}$$

Eqn 8.29

Upper case H

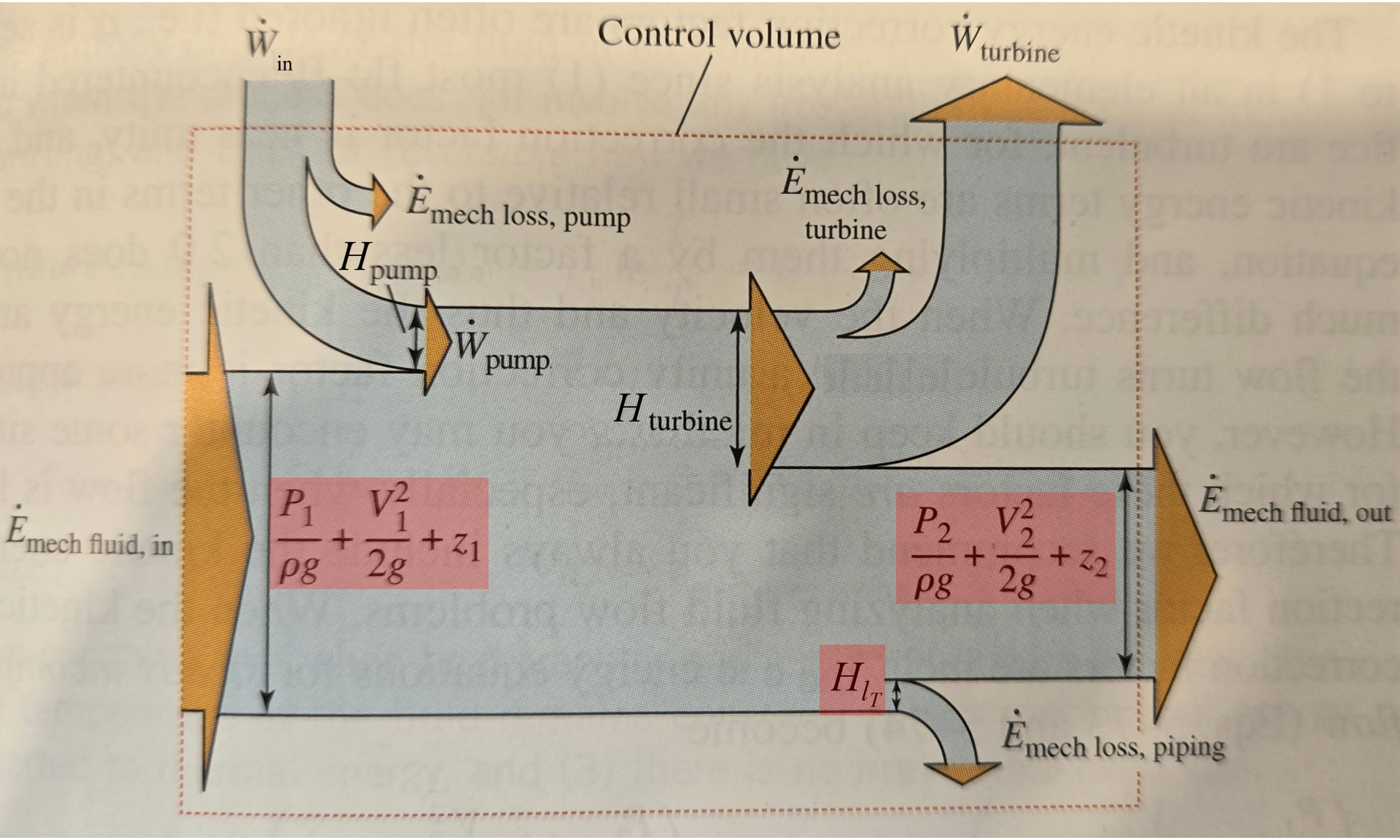
Units = length

$$\frac{P_1}{\rho g} + \alpha_1 \frac{V_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \alpha_2 \frac{V_2^2}{2g} + z_2 + H_{l_T}$$

Eqn 8.30

$$H_{l_T} = \frac{h_{l_T}}{g}$$

Sankey diagram of the full Bernoulli Equation



Frictional (i.e. major) losses

Pipe length

Avg. Velocity between points 1 and 2

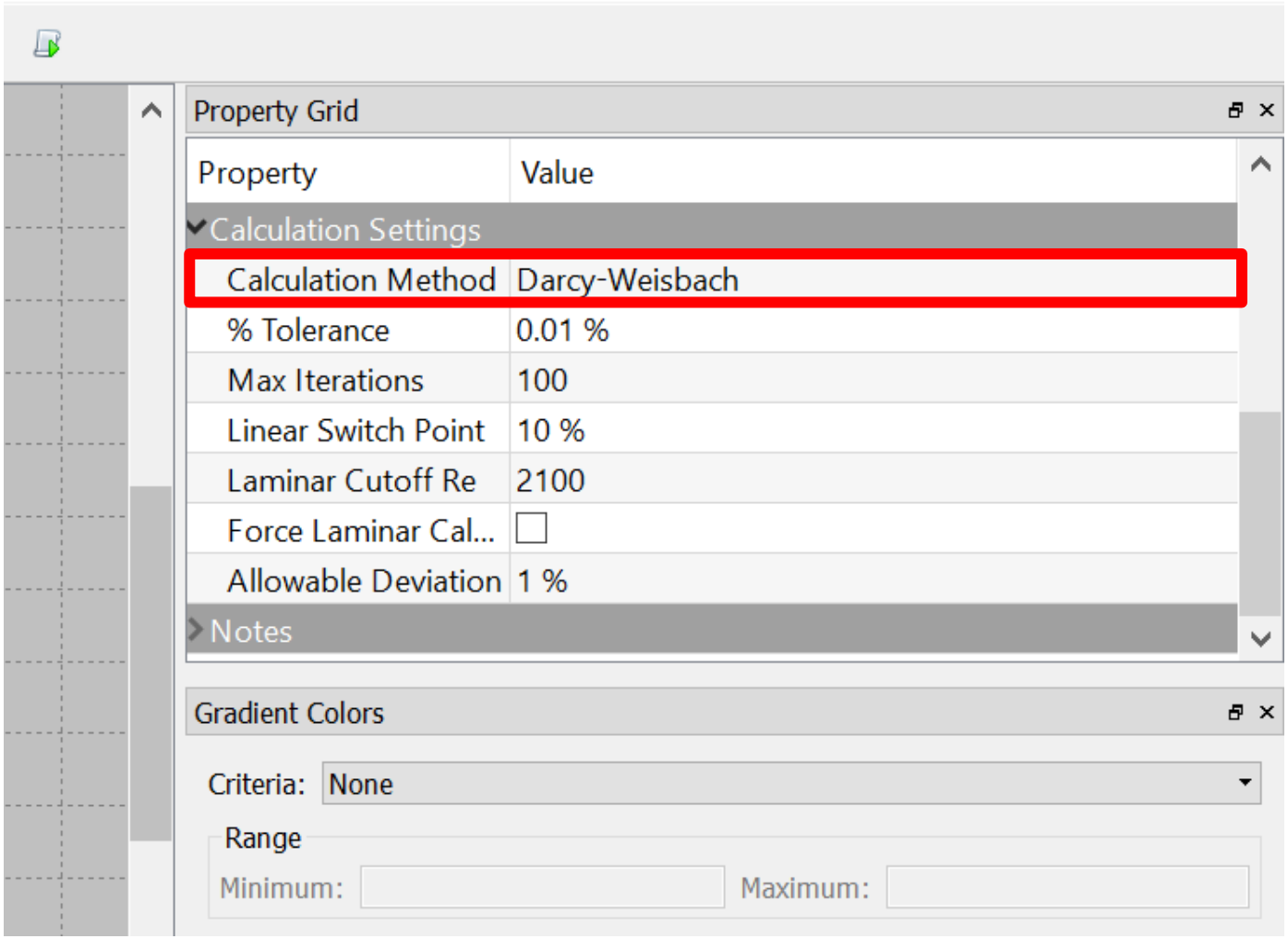
$$h_l = f \frac{L}{D_h} \frac{V_{avg}^2}{2}$$ Eqn 8.34

Darcy-Weisbach friction factor

Hydraulic diameter

$$H_l = f \frac{L}{D_h} \frac{V_{avg}^2}{2g}$$ Eqn 8.35

PIPE-FLO default settings



Frictional (i.e. major) losses

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Darcy-Weisbach friction factor

Hydraulic diameter

$$H_l = f \frac{L}{D_h} \frac{V_{avg}^2}{2g}$$ Eqn 8.35

$$D_h \equiv \frac{4A}{P}$$

Eqn 8.50

Cross-sectional area

'Wetted' perimeter (the portion of the duct perimeter that is physically touching the fluid)

$$\frac{P_1}{\rho} + \alpha_1 \frac{{V_1}^2}{2} + gz_1 = \frac{P_2}{\rho} + \alpha_2 \frac{{V_2}^2}{2} + gz_2 + h_l$$

- Need to calculate h_l

Need f

Need Re and ε

Need ρ, V_{avg}, D and μ

Need e and D

Look up in tables!

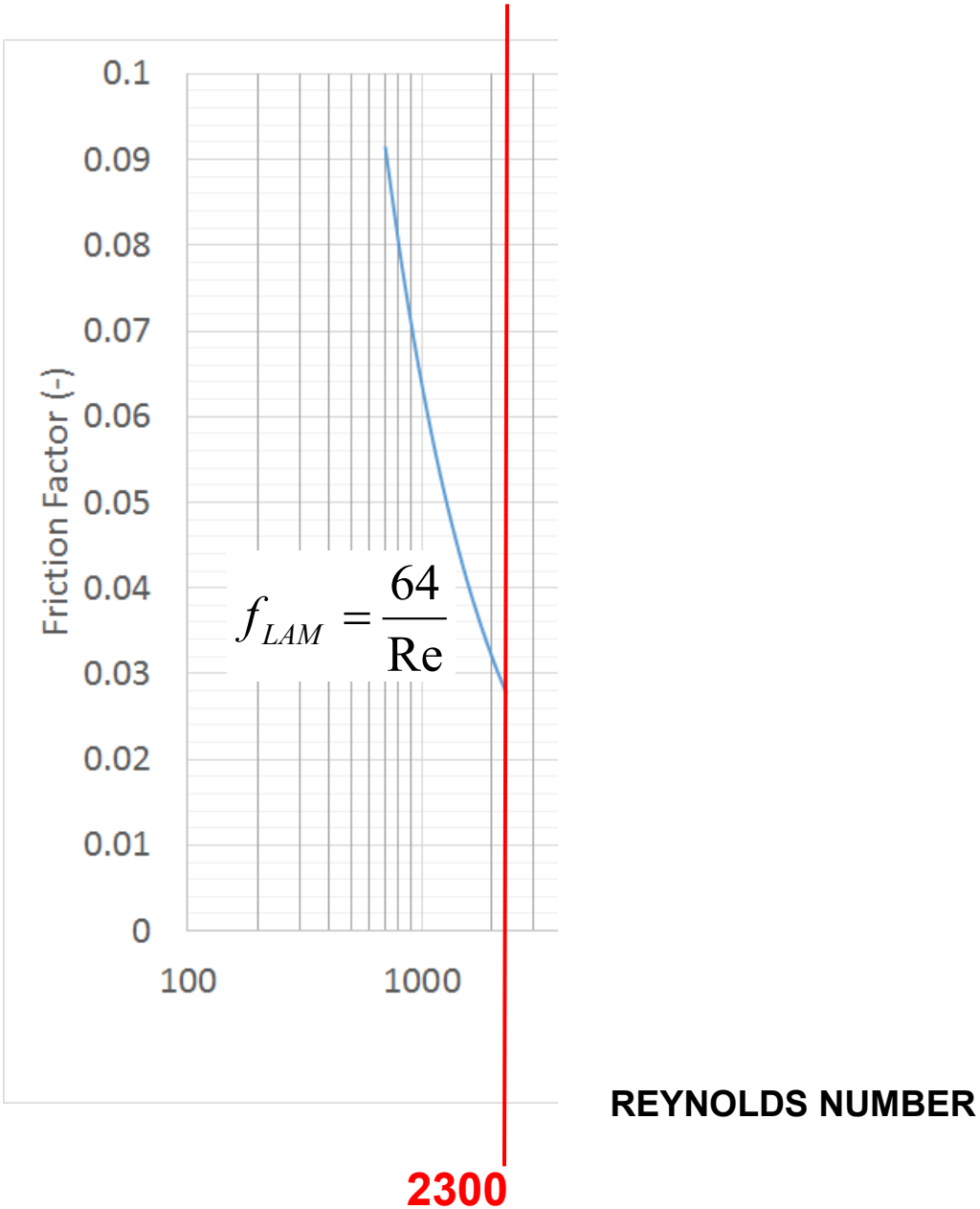
Laminar flow in a round pipe

- Laminar flow criteria → $Re = \frac{\rho V_{avg} D}{\mu} < 2300$

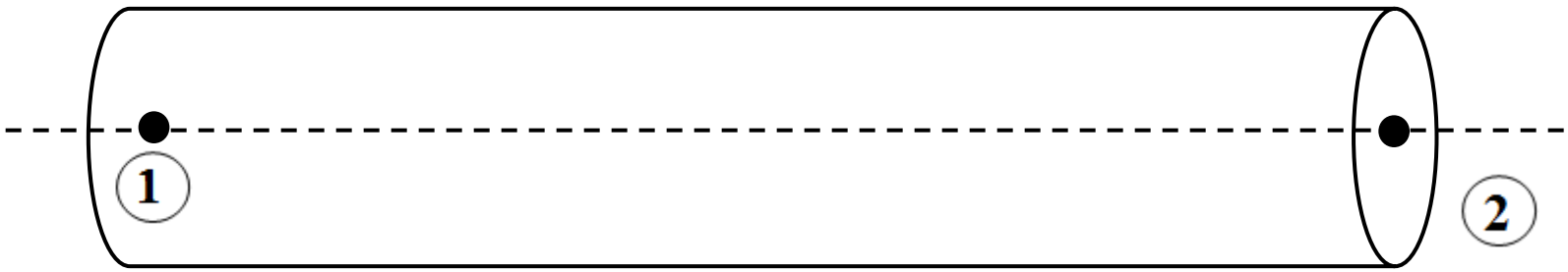
- Analytical solution available for Darcy-Weisbach friction factor:

$$f_{LAM} = \frac{64}{\mathrm{Re}}$$

Eqn 8.36

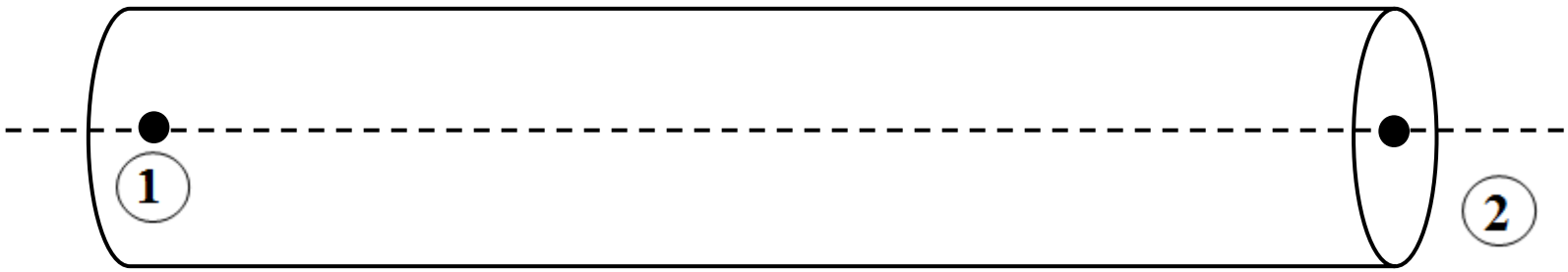

0.1
0.09
0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0
Friction Factor (-)
100
1000
$f_{LAM} = \frac{64}{\mathrm{Re}}$
2300
REYNOLDS NUMBER

Re-consider laminar flow through a horizontal pipe of constant diameter.



$$\frac{P_1}{\rho g} + \alpha_1 \frac{V_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \alpha_2 \frac{V_2^2}{2g} + z_2 + H_l + H_{l_m}$$

Re-consider laminar flow through a horizontal pipe of constant diameter.



$$P_1 - P_2 = \left(\frac{64\mu}{V_{avg}D}\right)\frac{L}{D}\frac{{V_{avg}}^2}{2}$$

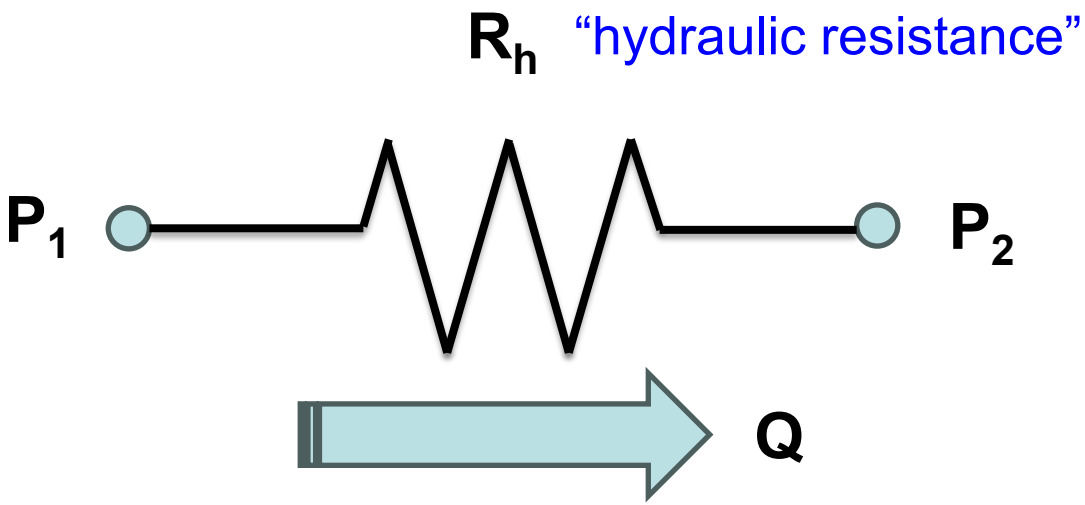
$$P_1 - P_2 = \frac{32\mu L}{D^2}V_{avg}$$

$$P_1 - P_2 = \frac{32\mu L}{(2R)^2}\frac{Q}{\pi R^2}$$

Hagen-Poiseuille Equation

$$\boxed{P_1 - P_2 = \frac{8\mu L Q}{\pi R^4}}$$

Analogy to electrical circuits

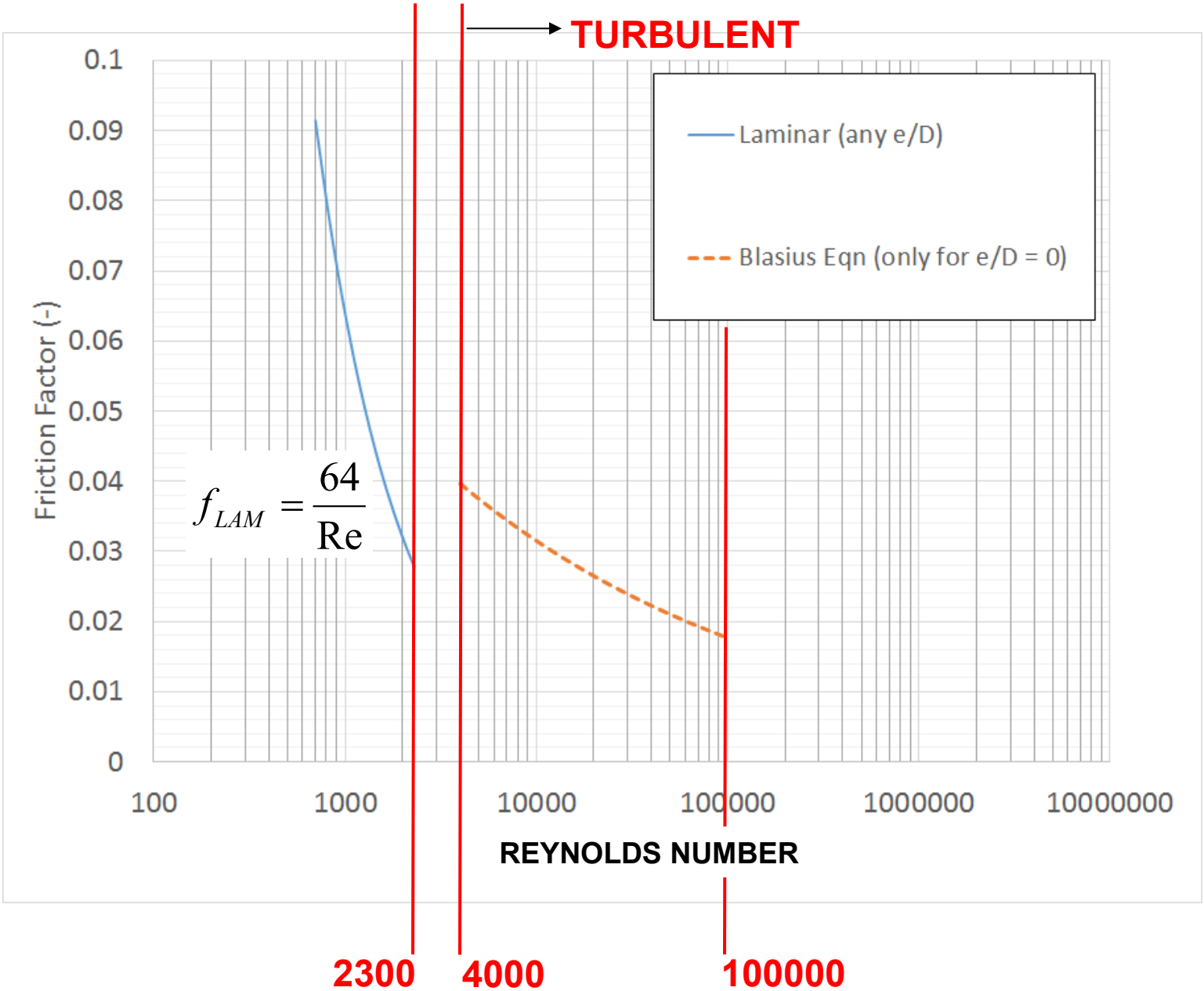


Turbulent flow in a round pipe

- Turbulent flow criteria → $Re = \dfrac{\rho V_{avg} D}{\mu} > 4000$
- Empirical expressions for Darcy-Weisbach friction factor:
  - Blasius Equation (smooth pipe, 4000 < Re < 100000 (i.e. 10^5)

$$f = \frac{0.316}{Re^{0.25}}$$

Eqn 8.38


TURBULENT
0.1
0.09
0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0
Friction Factor (-)
Laminar (any e/D)
Blasius Eqn (only for e/D = 0)
$f_{LAM} = \frac{64}{\mathrm{Re}}$
100
1000
10000
100000
1000000
10000000
REYNOLDS NUMBER
2300
4000
100000

Turbulent flow in a round pipe

- Turbulent flow criteria → $Re = \dfrac{\rho V_{avg} D}{\mu} > 4000$
- Empirical expressions for Darcy-Weisbach friction factor:
  - Blasius Equation (smooth pipe; 4000 < Re < 100000 (i.e. 10^5))

$$f = \frac{0.316}{Re^{0.25}}$$

Eqn 8.38

  - Colebrook Equation (any pipe, Re > 4000)

$$\frac{1}{\sqrt{f}} = -2.0 \log_{10}\left[\frac{\varepsilon}{3.7} + \frac{2.51}{\mathrm{Re}\sqrt{f}}\right]$$

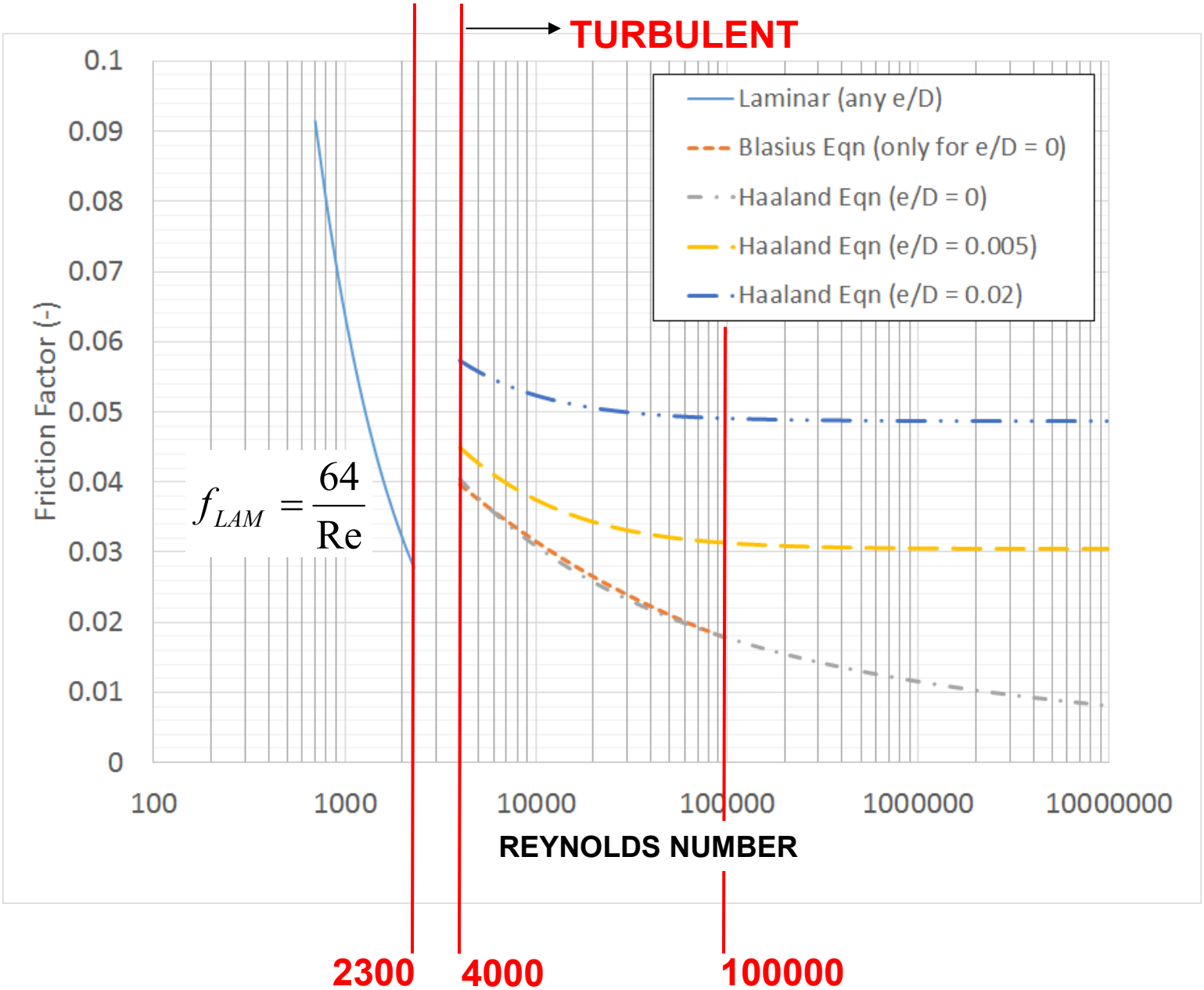
ε is relative roughness of the pipe (dimensionless)

Eqn 8.37a

  - Haaland Equation (any pipe, Re > 4000)

Eqn 8.37b

$$\frac{1}{\sqrt{f}} = -1.8 \log_{10}\left[\left(\frac{\varepsilon}{3.7}\right)^{1.11} + \frac{6.9}{\mathrm{Re}}\right] \Rightarrow \boxed{f = \left[\frac{1}{-1.8 \log_{10}\left[\left(\frac{\varepsilon}{3.7}\right)^{1.11} + \frac{6.9}{\mathrm{Re}}\right]}\right]^2}$$


TURBULENT
Laminar (any e/D)
Blasius Eqn (only for e/D = 0)
Haaland Eqn (e/D = 0)
Haaland Eqn (e/D = 0.005)
Haaland Eqn (e/D = 0.02)
Friction Factor (-)
0.1
0.09
0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0
100
1000
10000
100000
1000000
10000000
REYNOLDS NUMBER
$f_{LAM} = \frac{64}{\mathrm{Re}}$
2300
4000
100000

Turbulent flow in a round pipe

- PIPE-FLO uses Serghide equations to calculation the Darcy-Weisbach friction factor:

$$A = -2\log_{10}\left(\frac{\varepsilon}{3.7} + \frac{12}{\mathrm{Re}}\right)$$

$$B = -2\log_{10}\left(\frac{\varepsilon}{3.7} + \frac{2.51A}{\mathrm{Re}}\right)$$

$$C = -2\log_{10}\left(\frac{\varepsilon}{3.7} + \frac{2.51B}{\mathrm{Re}}\right)$$

$$f = \left(A - \frac{(B-A)^2}{C - 2B + A}\right)^{-2}$$

See pp 40-41 of PIPE-FLO Courseware

$$\frac{P_1}{\rho} + \alpha_1 \frac{{V_1}^2}{2} + gz_1 = \frac{P_2}{\rho} + \alpha_2 \frac{{V_2}^2}{2} + gz_2 + h_l$$

- Need to calculate h_l

Need f

Need Re and ε

Need ρ, V_{avg}, D and μ

Need e and D

Look up in tables!

Relative roughness $$\varepsilon = \frac{e}{D}$$

Pipe roughness [=] length

Pipe diameter [=] length

Table 8.1

Roughness for Pipes of Common Engineering Materials

	Roughness, e	
Pipe	Feet	Millimeters
Riveted steel	0.003–0.03	0.9–9
Concrete	0.001–0.01	0.3–3
Wood stave	0.0006–0.003	0.2–0.9
Cast iron	0.00085	0.26
Galvanized iron	0.0005	0.15
Asphalted cast iron	0.0004	0.12
Commercial steel or wrought iron	0.00015	0.046
Drawn tubing	0.000005	0.0015

Source: Data from Moody [8].

Relative roughness $\varepsilon = \dfrac{e}{D}$

Pipe roughness [=] length

Pipe diameter [=] length

Surface	Absolute Roughness	
	10^{-3} (m)	(feet)
Copper, Lead, Brass, Aluminum (new)	0.001 - 0.002	3.3 - 6.7 10^{-6}
PVC and Plastic Pipes	0.0015 - 0.007	0.5 - 2.33 10^{-5}
Epoxy, Vinyl Ester and Isophthalic pipe	0.005	1.7 10^{-5}
Stainless steel	0.015	5 10^{-5}
Steel commercial pipe	0.045 - 0.09	1.5 - 3 10^{-4}
Stretched steel	0.015	5 10^{-5}
Weld steel	0.045	1.5 10^{-4}
Galvanized steel	0.15	5 10^{-4}
Rusted steel (corrosion)	0.15 - 4	5 - 133 10^{-4}
New cast iron	0.25 - 0.8	8 - 27 10^{-4}
Worn cast iron	0.8 - 1.5	2.7 - 5 10^{-3}
Rusty cast iron	1.5 - 2.5	5 - 8.3 10^{-3}
Sheet or asphalted cast iron	0.01 - 0.015	3.33 - 5 10^{-5}
Smoothed cement	0.3	1 10^{-3}
Ordinary concrete	0.3 - 1	1 - 3.33 10^{-3}
Coarse concrete	0.3 - 5	1 - 16.7 10^{-3}
Well planed wood	0.18 - 0.9	6 - 30 10^{-4}
Ordinary wood	5	16.7 10^{-3}

http://www.engineeringtoolbox.com/major-loss-ducts-tubes-d_459.html

From Fluid Mechanics with Engineering Applications (Daugherty & Franzini):

"In planning for the future capacities it must be recalled that scale deposits will increase the roughness and reduce the cross-sectional area. For pipes in water service, the absolute roughness of old pipes (20 years and more) may increase over that of new pipes by threefold for concrete or cement-lined steel, up to twenty-fold for cast iron, and even to forty-fold for tuberculated wrought-iron and steel pipe."

Age Effects on Iron-Based Pipes in Water Distribution Systems

Laboratory testing was conducted in order to evaluate the hydraulic and physical characteristics of sections of aged pipe. Eleven aged pipe sections were obtained from water utilities and subjected to physical testing. The pipe sections varied in age from 25 to 50 years and in nominal diameter from 0.75-in to 4-in. Most were acquired when utilities were repairing or replacing a pipe section.

Table 2: Description of Aged Pipes

Pipe Section	Pipe Description	D (in)	d (in)	d/D	Percent Reduction in Flow Area
1	galvanized steel	0.824	0.767	0.931	13
2	galvanized steel	0.824	0.777	0.943	11
3	galvanized steel	0.824	0.781	0.948	10
4	galvanized steel	1.049	0.919	0.876	23
5	galvanized steel	1.049	0.958	0.914	16
6	galvanized steel	2.067	1.970	0.953	9.2

Pipe Properties

Name: Cast Iron SCH 20

Notes

Installed Pipe Tables

- Academic
- Aluminum B241
- Aluminum B307
- Aluminum B345
- Cast Iron Pipe
  - 20
  - 21
  - 22
  - 23
  - 24
  - 25
  - 26
  - 27
  - 28
  - 29
  - 30

Download Pipe Tables

Valve Table: standard

Pipe

Material: Cast Iron Pipe

Schedule: 20

Roughness: 0.2591 mm

C: 130

Size Range: 200 mm - 1220 mm

Sizing Criteria

Criteria: Design velocity

Velocity: 15 m/s

Design Limits

	min	max
Velocity (m/s)		
Pressure (Pa a)	0	1E+06
Reynolds No		

Handbook of Cast Iron Pipe - Version: 1

OK Cancel Help

Relative roughness $\varepsilon = \dfrac{e}{D}$

Pipe roughness [=] length

Pipe diameter [=] length

Table 8.5

Standard Sizes for Carbon Steel, Alloy Steel, and Stainless Steel Pipe

Nominal Pipe Size (in.)	Inside Diameter (in.)	Nominal Pipe Size (in.)	Inside Diameter (in.)
$\frac{1}{8}$	0.269	$2\frac{1}{2}$	2.469
$\frac{1}{4}$	0.364	3	3.068
$\frac{3}{8}$	0.493	4	4.026
$\frac{1}{2}$	0.622	5	5.047
$\frac{3}{4}$	0.824	6	6.065
1	1.049	8	7.981
$1\frac{1}{2}$	1.610	10	10.020
2	2.067	12	12.000

Source: Data from Reference [11].

See additional table of pipe specifications (from Janna textbook) posted on AVENUE

See additional table of pipe specifications (from Janna textbook) posted on AVENUE

TABLE D.1. *Pipe dimensions.*

Nominal Diameter	Outside Diameter in. (ft)	Outside Diameter cm	Schedule	Inside Diameter ft	Inside Diameter cm	Flow Area ft^2	Flow Area cm^2
1/8	0.405	1.029	40 (std)	0.02242	0.683	0.0003947	0.366 4
	(0.03375)		80 (xs)	0.01792	0.547	0.0002522	0.235 0
1/4	0.540	1.372	40 (std)	0.03033	0.924	0.0007227	0.670 6
	(0.045)		80 (xs)	0.02517	0.768	0.0004974	0.463 2
3/8	0.675	1.714	40 (std)	0.04108	1.252	0.001326	1.233
	(0.05625)		80 (xs)	0.03525	1.074	0.0009759	0.905 9
1/2	0.840	2.134	40 (std)	0.05183	1.580	0.002110	1.961
	(0.070)		80 (xs)	0.04550	1.386	0.001626	1.508
			160	0.03867	1.178	0.001174	1.090
			(xxs)	0.02100	0.640	0.0003464	0.321 7
3/4	1.050	2.667	40 (std)	0.06867	2.093	0.003703	3.441
	(0.0875)		80 (xs)	0.06183	1.883	0.003003	2.785
			160	0.05100	1.555	0.002043	1.898
			(xxs)	0.03617	1.103	0.001027	0.955 5
1	1.315	3.340	40 (std)	0.08742	2.664	0.006002	5.574
	(0.1095)		80 (xs)	0.07975	2.430	0.004995	5.083
			160	0.06792	2.070	0.003623	3.365
			(xxs)	0.04992	1.522	0.001957	1.815

TABLE D.1. continued. *Pipe dimensions.*

See additional table of pipe specifications (from Janna textbook) posted on AVENUE

Nominal Diameter	Outside Diameter in. (ft)	Outside Diameter cm	Schedule	Inside Diameter ft	Inside Diameter cm	Flow Area ft^2	Flow Area cm^2
2	2.375	6.034	40 (std)	0.1723	5.252	0.02330	21.66
	(0.1979)		80 (xs)	0.1616	4.926	0.02051	19.06
			160	0.1406	4.286	0.01552	14.43
			(xxs)	0.1253	3.820	0.01232	11.46
$2^1/_2$	2.875	7.303	40 (std)	0.2058	6.271	0.03325	30.89
	(0.2396)		80 (xs)	0.1936	5.901	0.02943	27.35
			160	0.1771	5.397	0.02463	22.88
			(xxs)	0.1476	4.499	0.01711	15.90
3	3.500	8.890	40 (std)	0.2557	7.792	0.05134	47.69
	(0.2917)		80 (xs)	0.2417	7.366	0.04587	42.61
			160	0.2187	6.664	0.03755	34.88
			(xxs)	0.1917	5.842	0.02885	26.80
$3^1/_2$	4.000	10.16	40 (std)	0.2957	9.012	0.06866	63.79
	(0.3333)		80 (xs)	0.2803	8.544	0.06172	57.33
4	4.500	11.43	40 (std)	0.3355	10.23	0.08841	82.19
	(0.375)		80 (xs)	0.3198	9.718	0.07984	74.17
			120	0.3020	9.204	0.07163	66.54
			160	0.2865	8.732	0.06447	59.88
			(xxs)	0.2626	8.006	0.05419	50.34
5	5.563	14.13	40 (std)	0.4206	12.82	0.1389	129.10
	(0.4636)		80 (xs)	0.4011	12.22	0.1263	117.30
			120	0.3803	11.59	0.1136	105.50
			160	0.3594	10.95	0.1015	94.17
			(xxs)	0.3386	10.32	0.09004	83.65

Friction factor

- Laminar:

$$f_{LAM} = \frac{64}{\mathrm{Re}}$$ (any pipe, Re < 2300)

- Turbulent:
  - Blasius Equation (smooth pipe, 4000 < Re < 100000)
  - Colebrook Equation (any pipe, Re > 4000)
  - Haaland Equation (any pipe, Re > 4000)
  - Serghide Equations (any pipe, Re > 4000)

- Moody Chart (proposed in 1944)

Friction Factors for Pipe Flow

By LEWIS F. MOODY,[1] PRINCETON, N. J.

The object of this paper is to furnish the engineer with a simple means of estimating the friction factors to be used in computing the loss of head in clean new pipes and in closed conduits running full with steady flow. The modern developments in the application of theoretical hydrodynamics to the fluid-friction problem are impressive and scattered through an extensive literature. This paper is not intended as a critical survey of this wide field. For a concise review, Professor Bakhmeteff's (1)[2] small book on the mechanics of fluid flow is an excellent reference. Prandtl and Tietjens (2) and Rouse (3) have also made notable contributions to the subject. The author does not claim to offer anything particularly new or original, his aim merely being to embody the now accepted conclusions in convenient form for engineering use.

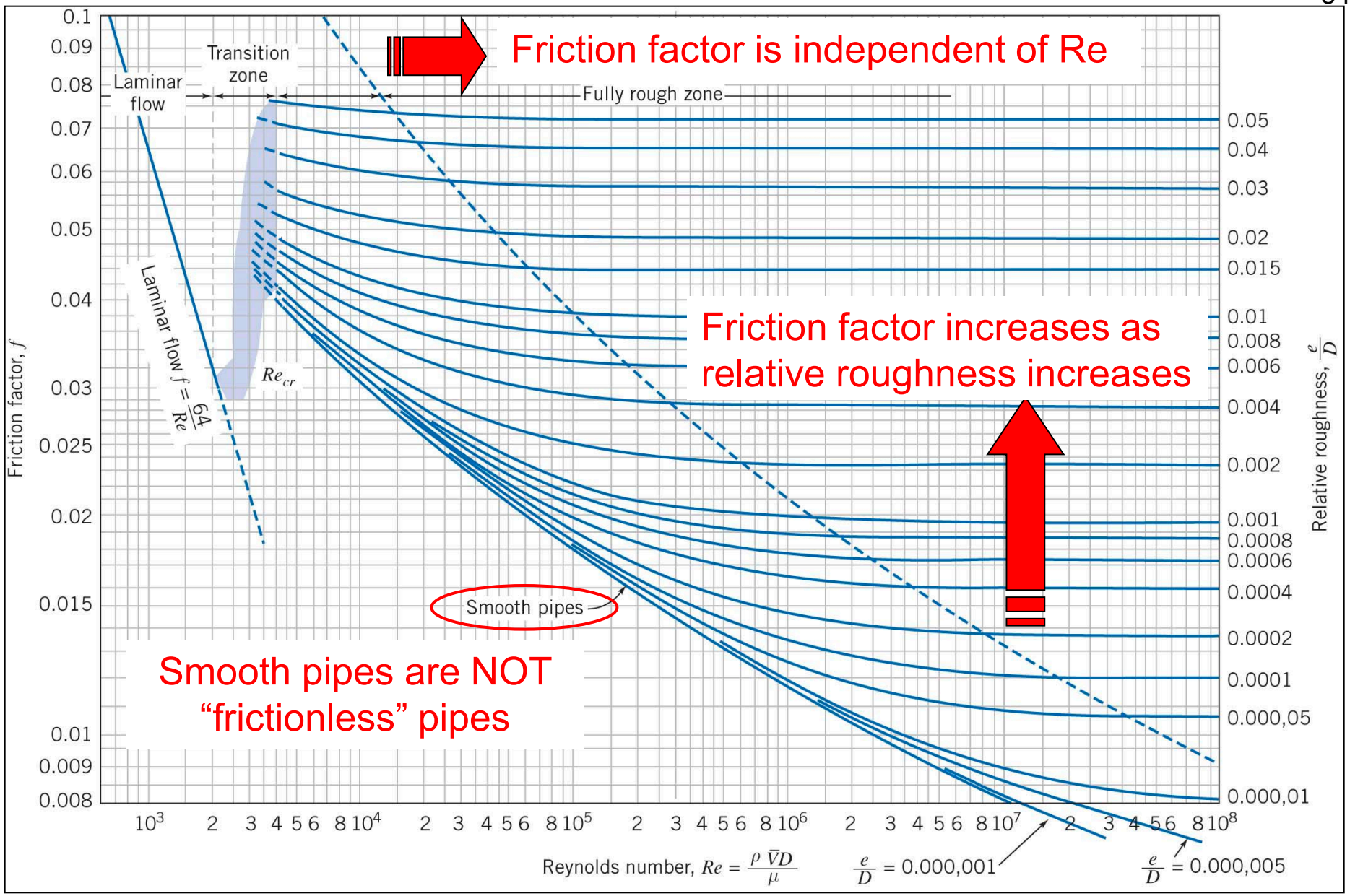
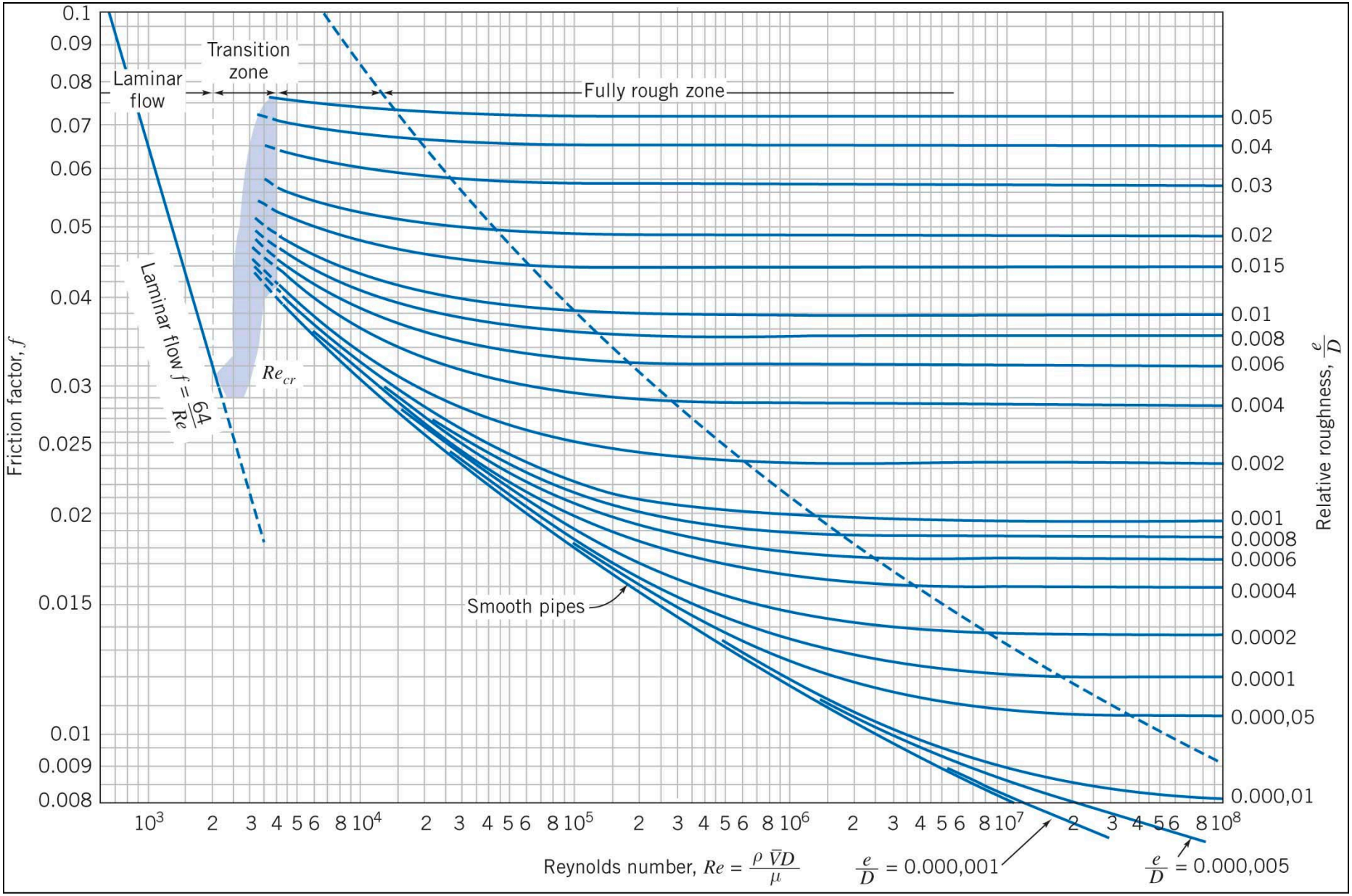
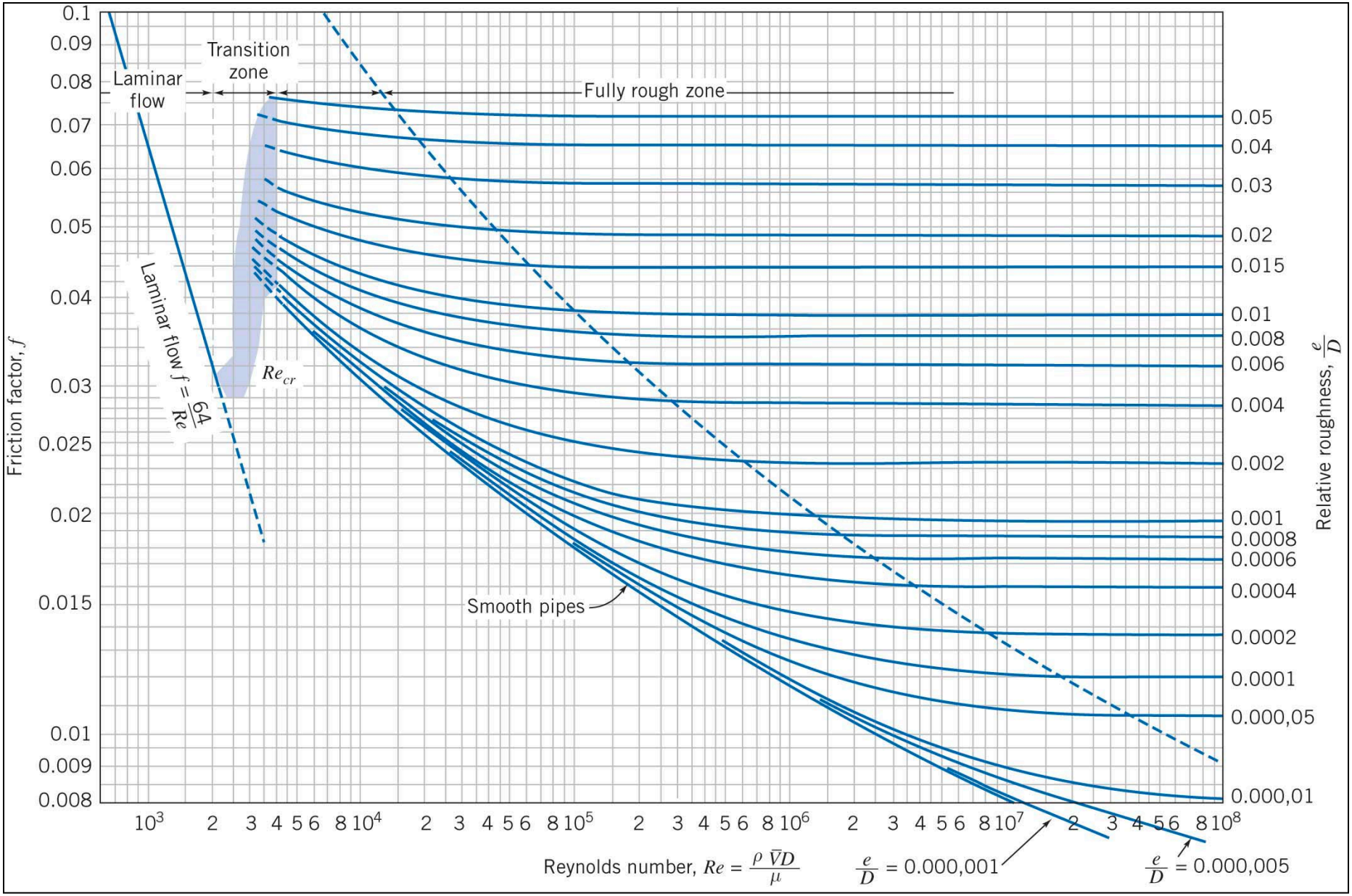

Friction factor is independent of Re
Laminar flow
Transition zone
Fully rough zone
Laminar flow $f = \frac{64}{Re}$
Re_{cr}
Friction factor increases as relative roughness increases
Smooth pipes
Smooth pipes are NOT "frictionless" pipes
Friction factor, f
Relative roughness, $\frac{e}{D}$
Reynolds number, $Re = \frac{\rho \bar{V} D}{\mu}$
$\frac{e}{D} = 0.000,001$
$\frac{e}{D} = 0.000,005$
0.1
0.09
0.08
0.07
0.06
0.05
0.04
0.03
0.025
0.02
0.015
0.01
0.009
0.008
10^3 2 3 4 5 6 8 10^4 2 3 4 5 6 8 10^5 2 3 4 5 6 8 10^6 2 3 4 5 6 8 10^7 2 3 4 5 6 8 10^8
0.05
0.04
0.03
0.02
0.015
0.01
0.008
0.006
0.004
0.002
0.001
0.0008
0.0006
0.0004
0.0002
0.0001
0.000,05
0.000,01

Figure 8.13

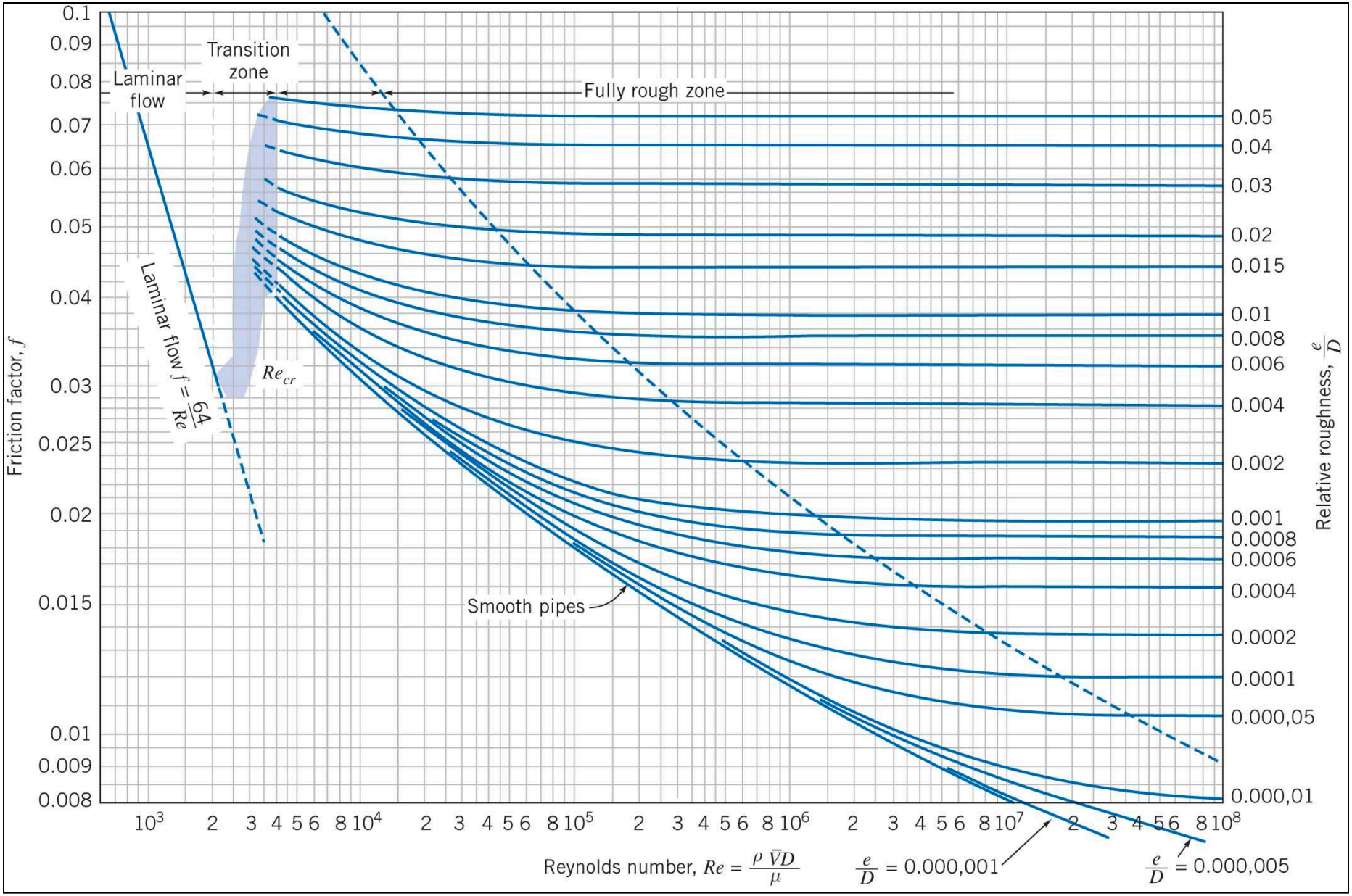
- For Re = 25 000 and ε = 0.002, f = _____



- For f = 0.012 and 'smooth pipe', Re = _____

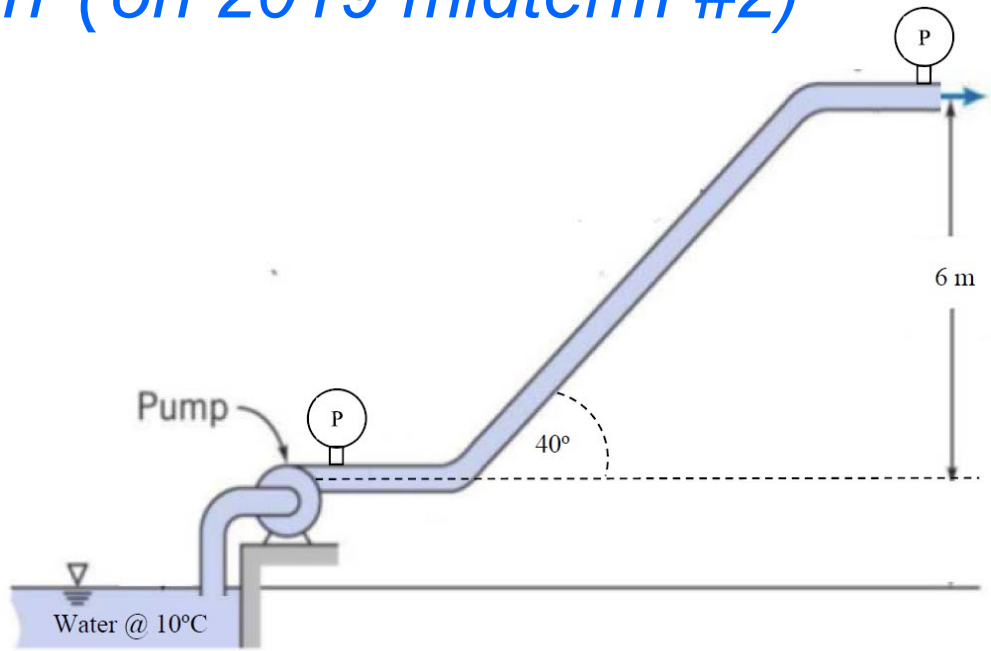


- For Re = 1000 and ε = 0.02, f = _____



In-class problem (on 2019 midterm #2)

A new pump-piping system is to be designed to deliver 500 L/min of water (at 10 °C) uphill from a large reservoir to a construction site. As shown in the figure below, the upper portion of the piping system is 6 m above the lower portion and the angle of the inclined section of pipe is 40° relative to the horizon. The same type of piping (2½" schedule 40 cast iron) is used for the entire length on the discharge side of the pump.



a. Write the simplified form of the Bernoulli equation for the condition where reference points 1 & 2 are chosen to be where the two pressure gauges are located on the discharge side of the pump. NOTE: you can ignore the minor losses associated with the two elbows that are needed to make the incline.

b. If the differential pressure reading between the two pressure gauges is reported to be 84.5 kPa, determine the total length of the 2½" schedule 40 cast iron piping between the two pressure gauges.

c. With extended use, it is expected that the cast iron pipe will become worn and rusty. Comment on which variable in the Bernoulli equation would account for this change and what is the overall effect on the performance of the pump-piping system. Give a BRIEF explanation (no calculations are needed!) of how you would determine the actual change in the performance of the pump-piping system.

Problems from textbook for self-study

10th Edition	9th Edition
6.12	6.36
6.13	6.30
6.14	6.38
6.16	6.42
6.17	6.43
8.1	8.1
8.2	8.2
8.14	8.37
8.16	8.39
8.25	8.60

PIPE-FLO

Complete tutorial #4 'Simulation Calculations'